

ANTIBACTERIAL ACTIVITY OF SOME DIURETIC INDIGENOUS MEDICINAL PLANTS

MARYAM MIRZA*, ZAHRA YAQEEN, SHAKIRA QUADRI,
R.B. QADRI AND NAJMA MURTAZA
PCSIR Laboratories Complex, Karachi-75280

ABSTRACT

Zea mays (Corn silk) alcoholic extract showed greater activity against *Klebsiella pneumoniae*, *Escherichia coli*, *Salmonella paratyphi* A and *Salmonella typhi* as compare to *Penbritin*, while *Raphanus sativus* Linn (Seeds) alcoholic extract showed greater activity against all the three organism except *Klebsiella pneumoniae* (gram -ve) as compare to *Penbritin*. *Raphanus sativus* Linn (Seeds) alcoholic extract and *Zea mays* (Corn silk) alcoholic extract showed greater activity against *Bacillus subtilis* (gram +ve) in comparison with *Penbritin*, *Cymbopogon citratus* (DC) Stapf leaves powder and *Raphanus sativus* Linn (seeds) oil did not show any antibacterial activity. *Raphanus sativus* Linn (seeds) oil did not show any antifungal activity.

INTRODUCTION

Cymbopogon citratus (DC) Stapf:

Cymbopogon citratus (DC) Stapf (Poaceae) commonly known as “serai” lemon grass, Family Gramineae (Maryam *et al.*, 2001, Chopra *et al.*, 1958). Graminae, one of the largest family of the flowering plants, comprising of 620 genera and C. 1000 species is widely distributed throughout the world. In Pakistan, this family is represented by 158 genera and 492 species (Cope, 1982 and Stewart, 1972). Lemon grass which is a perennial grass, also possesses a C₄ photosynthetic pathway (NADP-ME type) (Hatch *et al.*, 1970).

Lemon grass is reported to have many medicinal properties. It is used as sudorific, stimulant, antiperiodic and anticatarrhal (Chopra *et al.*, 1956). A decoction made from the leaves is recommended as diaphoretic in fever (Chopra *et al.*, 1958).

An infusion of lemon grass is sometimes taken as refreshing beverage and this use gives it a local name “Hirvacha” or green tea

(Anonymous, 1950). In Java, it is used in preparation of highly spiced “sherbet” (Burkill, 1935). A number of studies are available regarding the antimicrobial activity (Ibrahim, 1992) and antimutagenic properties of the plant extract towards chemically induced mutation in *Salmonella typhimurium* strains TA 98 and TA 100 (Vinitketumnuen *et al.*, 1994).

Raphanus sativus Linn:

Raphanus sativus Linn commonly known as radish (Muli safe) belongs to family Cruciferae and cultivated throughout subcontinent Indo-Pak in gardens and plains for culinary purpose (Nadkarni, 1954). The plant is an excellent source of vitamins B and C. Particularly in India this plant is used as purgative, stimulant, antisorbetic, diuretic and lithotryptic. The seeds have been used as emmenagogue and in treatment of gonorrhea and cancer (Watt John Mitchell *et al.*, 1962).

Root is a reputed medicine for piles, gastrodynic, pains, urinary and syphilitic complaints, relieve dysuria and strangury (Chopra *et al.*, 1958 and Nadkarni, 1954). Radish root eaten before meal improves appetite and increases the digestive powder,

*Author for correspondence

while young radish (pods) is a diet for fistula in ano when there is no fever (Nadkarni, 1954).

Seeds of radish are expectorant, diuretic, laxative and carminative (Kirtikar *et al.*, 1933 and Chopra *et al.*, 1958).

Roots contain glycosides, enzyme and methyl mercaptan (Watt John Mitchell *et al.*, 1962). Fresh vegetable contains 91.00% moisture and seeds on extraction with petroleum ether yield 4.00% albuminoids 18.00%, soluble carbohydrates 52.66% woody fiber 9.34% and ash 16.00% (Nadkarni, 1954). Seed and root contains a fixed oil, essential oil, a sulphuretted volatile oil which resembles mustard seed oil. This oil contains sulfur and phosphoric acids (Nadkarni, 1954 and Watt John Mitchell *et al.*, 1962).

The seeds contain 300% fixed oil. The volatile oil is a sulfur oil. The seeds also contain sulfuric acid and erucic acid (Watt John Mitchell *et al.*, 1962).

***Zea mays* Linn:**

Zea mays belong to family Gramineae. Common name is 'Bhutta' (Chopra *et al.*, 1956). In English Maize, Indian Corn, it is locally called Makai (Dymock, 1890; Watt George, 1892 and Nadkarni, 1954). *Zea* consists of the fresh or dried stigmas and styles obtained from *Zea mays* L. The styles and stigmas should be collected from the unripe corn. *Zea* (Corn silk) contains a volatile, alkaloid, resins, maizeric acid, fixed oil and sugar. A yellow powder of styles consists of parenchyma with two vascular bundles composed of narrow annular or spiral tracheids, epidermal cells rectangular, often extended into multicellular hairs 0.2-0.8 mm. in length, the basal portion consisting of two to five united cells, the upper portion usually unicellular. Purple red parenchyma cells contain a red cell sap (British Herbal Pharmacopoeia, 1983 and Nadkarni, 1954).

The silky stigmata (Corn silk) are used in decoction in diseases of the bladder and

kidneys. It has marked diuretic action. It is given in lithiasis (Dymock 1890, Kirtikar *et al.*, 1933, Martin *et al.*, 1991, Watt George 1892, Youngken 1948 and Watt John Mitchell *et al.*, 1962). Corn silk contains 2% of maizenic acid, a fixed oil (*Oleum maydis*), resin, sugar, mucilage, salts and phlobaphene (Watt John Mitchell *et al.*, 1962). Corn silk (styles with stigma) is a chemical complex medicinal material having valuable properties. It has sitosterol, stigmasterol, fatty and volatile oil, saponin, a bitter glucoside substance, vitamin C, vitamin K., Corn silk is used as a choleric and diuretic agents and as an astringent (Bobryshev, 1962).

MATERIAL AND METHODS

Cymbopogon citratus (DC) Stapf leaves (Voucher No.499), *Raphanus sativus* Linn, seeds (Voucher No.501) and of *Zea mays* (Corn silk) (Voucher No.502); Seeds of above mention medicinal plants and *Zea mays* (Corn silk); one kg of each, were purchased from local market and *cymbopogon citratus* (DC) Stapf cultivated in PCSIR Laboratories Complex, Karachi and identified by the Pharmacognosy Section of these Laboratories.

1. Preparation of ethanolic extract:

Seeds and Corn silk of above mention medicinal plants were purchased from local market and washed thoroughly with tap water and then with distilled water and dried in open air at room temperature 37°C for 24 hours. Five Kg of each above mention, seeds and corn silk were chopped/grinded into a coarse powder to a kitchen chopper (Moulinex) and soaked 95% ethyl alcohol (3 litres) separately stirring was done by means of a mechanical stirrer (Gallen Kamp 280-010, Made in England) for one hour for one week. After one week, all parts of material filtered separately filtrate were evaporated reduce pressure using rotary evaporator (Büchi rota vapore El-131) at 40°C (Izhar *et al.*, 1988 and Siddiqui *et al.*, 1968). The resulting brownish yellow mass was obtained (2.50 kg.) (2.45 kg.) (2.65 kg.) (2.45 kg.).

Table-1
Antibacterial Activity Against Gram -ve and Gram +ve Bacteria
(Zone of Inhibition in mm)

Culture Used	Concentration mg/ml	Alcoholic Extract		Penbritin	Erythrocin	Ampiclox
		<i>Zea mays</i> Linn (corn silk)	<i>Raphanus</i> <i>sativus</i> Linn (Seeds)			
Gram -ve: <i>Klebsiella</i> <i>Pneumoniae</i>	10	20	10	15	17	16
	05	19	10	13	15	15
	02.5	10	10	12	13	13
	01.25	10	10	10	11	11
	control	10	10	10	10	10
<i>Escherichia coli</i>	10	20	19	15	24	21
	05	18	17	14	23	20
	02.5	17	16	13	22	18
	01.25	10	10	12	17	15
	control	10	10	10	10	10
<i>Salmonella</i> <i>paratyphi</i> A	10	21	22	21	15	23
	05	20	21	16	14	20
	02.5	10	20	13	13	18
	01.25	10	10	12	12	12
	control	10	10	10	10	10
<i>Salmonella</i> <i>typhi</i>	10	18	20	23	24	21
	05	16	18	20	22	19
	02.5	10	16	18	20	17
	01.25	10	10	16	16	16
	control	-ve	-ve	10	10	10
<i>Staphylococcus</i> <i>aureus</i>	10	-ve	-ve	14	24	22
	05	-ve	-ve	13	24	12
	02.5	-ve	-ve	12	23	17
	01.25	-ve	-ve	12	22	16
	control	-ve	-ve	10	10	10
<i>Streptococcus</i> <i>pyogenes</i>	10	-ve	21	21	29	34
	05	-ve	20	18	25	32
	02.5	-ve	19	15	24	30
	01.25	-ve	10	13	21	30
	control	-ve	10	10	10	10
<i>Lactobacillus</i> <i>plantarum</i>	10	13	16	17	27	22
	05	12	15	14	26	20
	02.5	10	14	13	24	19
	01.25	10	13	12	23	19
	control	10	10	10	10	10
<i>Bacillus</i> <i>subtilis</i>	10	23	21	14	24	21
	05	21	19	12	22	18
	02.5	19	18	11	20	17
	01.25	18	18	11	18	15
	control	10	10	10	10	10
<i>Bacillus</i> <i>pumilus</i>	10	15	14	15	25	23
	05	13	13	13	22	21
	02.5	10	12	12	22	20
	01.25	10	10	12	20	17
	control	10	10	10	10	10

(-ve) = No inhibition

Table-2

Antibacterial activity against pathogens zone of inhibition in mm.
Cymbopogon citratus (DC) Stapf leaves powder

Concentration of leaves powder mg/ml	<i>E. coli</i> gram -ve	<i>B. subtilis</i> gram +ve	<i>S. paratyphi</i> A gram -ve	<i>S. aureus</i> gram +ve	Ampiclox
10	-ve	-ve	-ve	-ve	22
5	-ve	-ve	-ve	-ve	18
2.5	-ve	-ve	-ve	-ve	14
1.25	-ve	-ve	-ve	-ve	12
Control	10	10	10	10	10

(-ve) = No inhibition

Table-3

Antibacterial activity against pathogens zone of inhibition in mm.
Raphanus sativus Linn seeds oil

Concentration of oil mg/ml	<i>E. coli</i> gram -ve	<i>B. subtilis</i> gram +ve	<i>S. paratyphi</i> B gram -ve	<i>S. aureus</i> gram +ve	Ampiclox
10	-ve	-ve	-ve	-ve	20
5	-ve	-ve	-ve	-ve	17
2.5	-ve	-ve	-ve	-ve	16
1.25	-ve	-ve	-ve	-ve	13
Control	10	10	10	10	10

Standard oil 10% Complex check.

(-ve) = No inhibition

Table-4

Antifungal activity of *Raphanus sativus* oil (zone of inhibition in mm)

Concentration mg/ml.	Antifungal Activity	
	<i>Aspergillus niger</i>	<i>Fusarium</i> sp.
5	-ve	-ve
2.5	-ve	-ve
1.25	-ve	-ve
Standard Tecto 5%	10	10
Control	10	10

(-ve) = No inhibition

2. Extraction of Oil:

The wash and dried seeds (50 gm) were subjected to a kitchen chopper (10,000 rpm) Moulinex [Model Depose – Brevets S.G.D.G. Pent Pending – Made in France]. Oil from the

seeds powder extracted in a soxhlet extractor with pet-ether (B.P. 40–60°C) 200 ml. The solvent was removed under reduced pressure to furnish a oil was golden yellow colour (14.87gms, 29.74%) (Zahra et al., 1999).

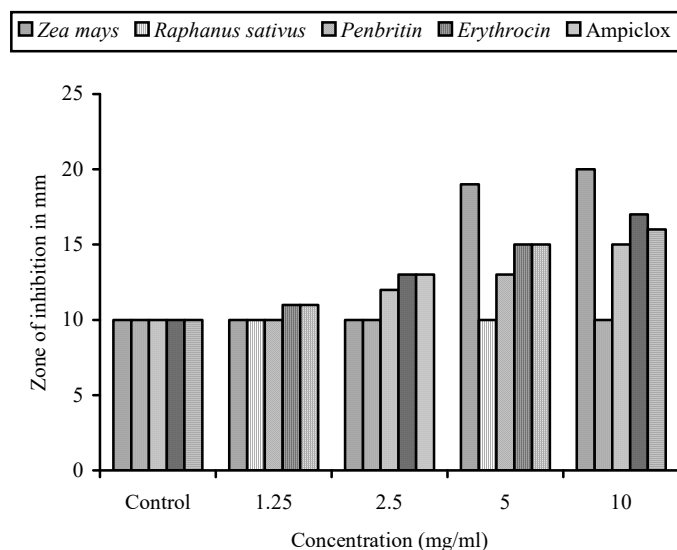


Fig. 1: Antibacterial activity against gram –ve bacteria *Klebsiella pneumoniae*.

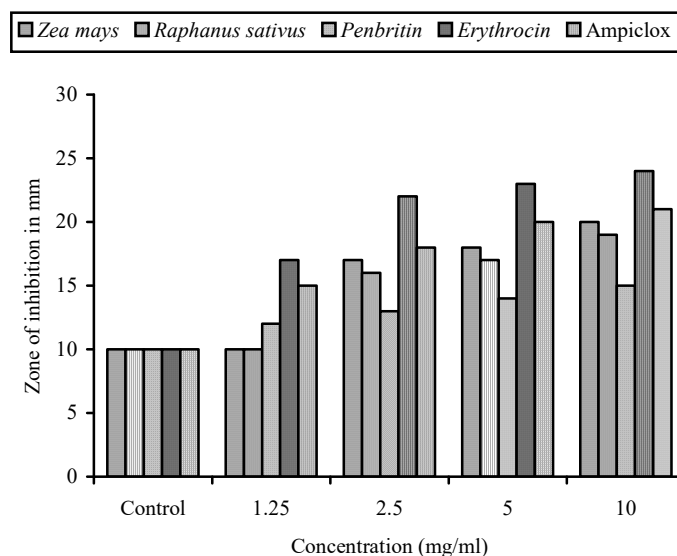


Fig. 2: Antibacterial activity against gram –ve bacteria *Escherichia coli*.

Zea mays Linn. (corn silk) and *Raphanus sativus* Linn. (seeds) were used in alcoholic extract.

Studies on the antibacterial activity of *zea mays* linn (corn silk) and *Raphanus sativus* linn (seeds) in ethanolic extract:

The studies on antibacterial activity/antifungal activity of plant extracts

present a unique challenge to strike new sources of medicine. Existing antibiotics have numerous drawbacks whereas the drugs of plant origin have no side effect or only marginal.

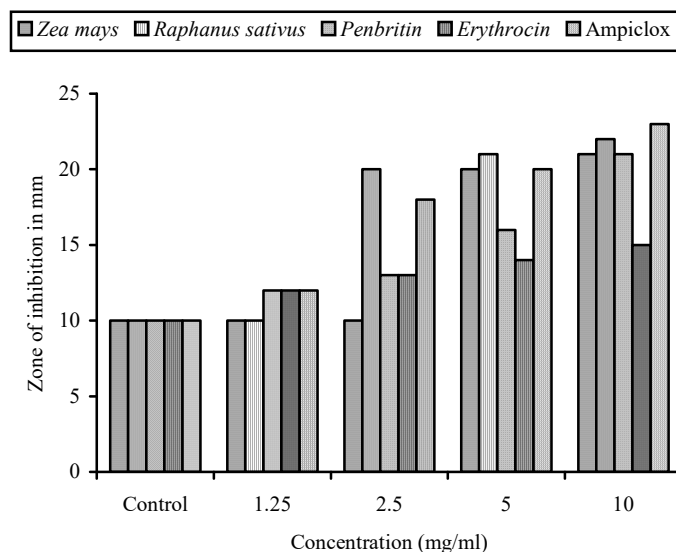


Fig. 3: Antibacterial activity against gram –ve bacteria *Salmonella paratyphi A*.

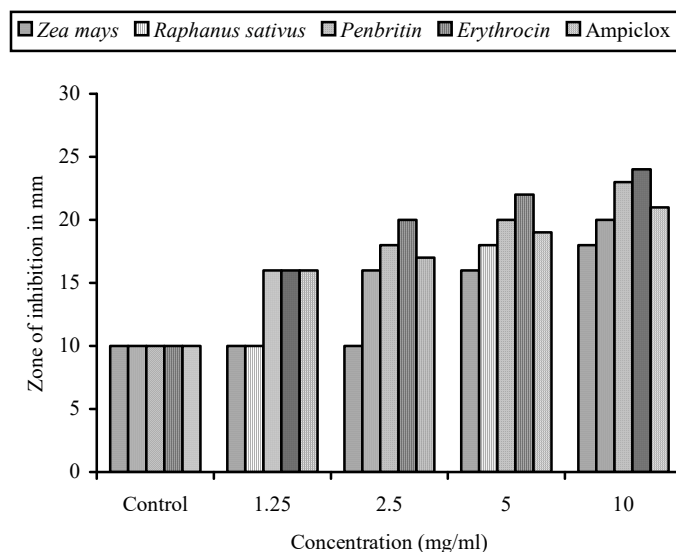


Fig. 4: Antibacterial activity against gram –ve bacteria *Salmonella typhi*.

Zea mays Linn. (corn silk) and *Raphanus sativus* Linn. (seeds) were used in alcoholic extract.

During the present studies medicinal value of diuretic plants *Cymbopogon citratus* (DC) Stapf (leaves), *Raphanus sativus* Linn (seeds) and *Zea mays* Linn (corn silk) have been determined by evaluating their antibacterial activity.

Antibacterial Activity

Cultures of bacteria employed during this study were four gram -ve, i.e., *Klebsiella pneumoniae*, *Escherichia coli*, *Salmonella paratyphi* A and *Salmonella paratyphi* and five gram +ve, i.e., *Staphylococcus aureus*,

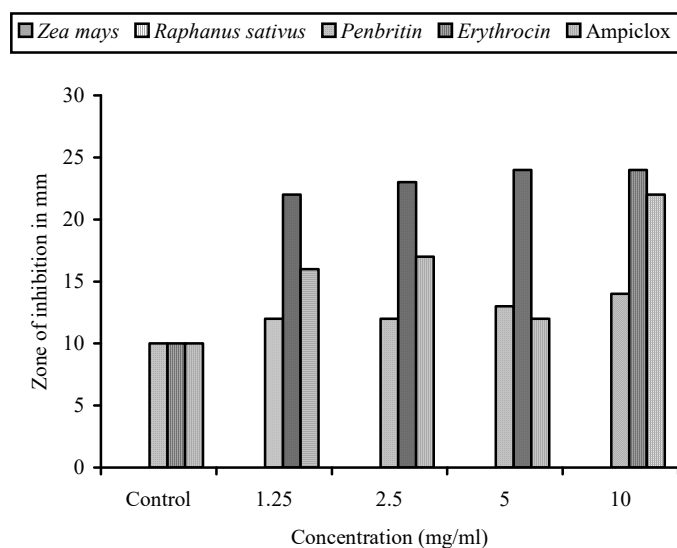


Fig. 5: Antibacterial activity against gram -ve bacteria *Staphylococcus aureus*.

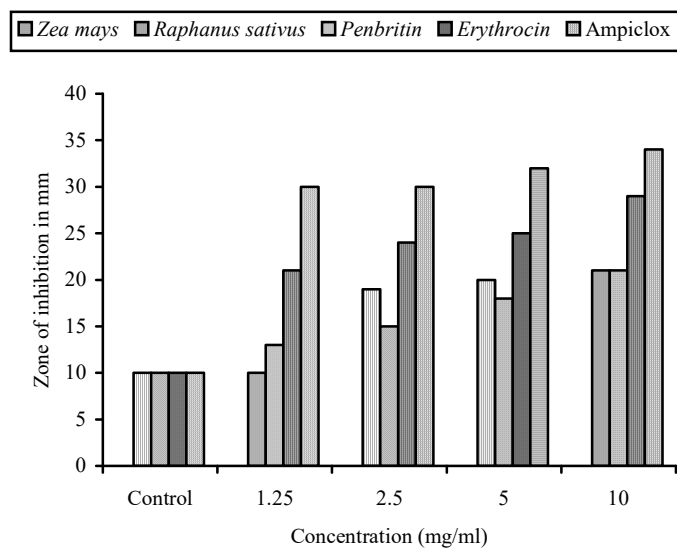


Fig. 6: Antibacterial activity against gram -ve bacteria *Staphylococcus pyogenes*.

Zea mays Linn. (corn silk) and *Raphanus sativus* Linn. (seeds) were used in alcoholic extract.

Streptococcus pyogenes, *Lactobacillus plantarum*, *Bacillus subtilis* and *Bacillus pumilus*.

All bacteria used in these tests were

inoculated on nutrient agar slants and incubated at 37°C for 24 hours. The 24 hours old cultures were grown on nutrient broth tubes (1 ml in each tube) and further incubated at 37°C for 24hrs. The well method (British

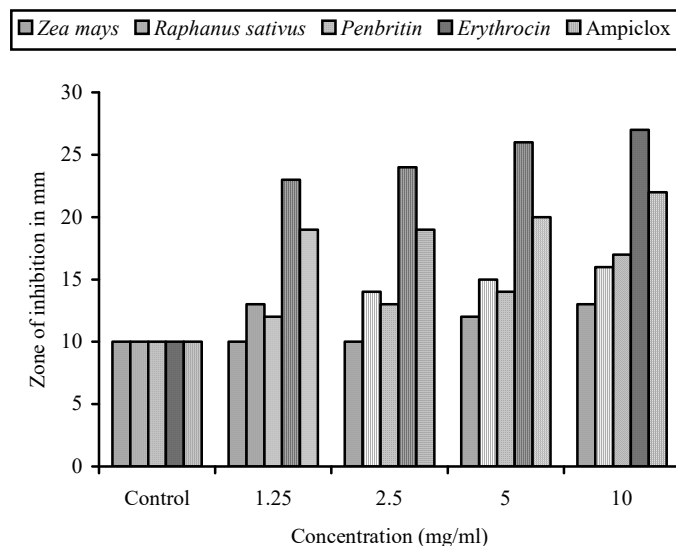


Fig. 7: Antibacterial activity against gram +ve bacteria *Lactobacillus plantarum*.

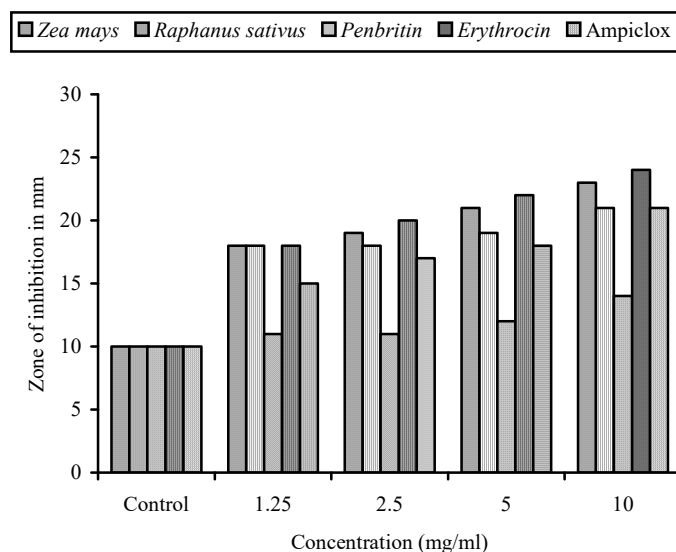


Fig. 8: Antibacterial activity against gram +ve bacteria *Bacillus subtilis*.

Zea mays Linn. (corn silk) and *Raphanus sativus* Linn. (seeds) were used in alcoholic extract.

Pharmacopoeia, 1980) was employed for testing antibacterial activity of the *Zea mays* Linn (corn silk) and *Raphanus sativus* (seeds) alcoholic extract against nine bacteria (four gram -ve and five gram +ve). The zones of inhibition were measure from the centre of the boarer to four different points on the margin of the zone. Three antibiotics, namely Erythrocin, Ampiclox and Penbritin were used for comparison (Table-1) (Najma Murtaza et al., 1994).

Antibacterial activity of *Cymbopogon citratus* (DC) Stapf leaves powder:

Antibacterial activity of *Cymbopogon citratus* (DC), Stapf leaves powder was tested against gram +ve and gram -ve bacteria. No antibacterial activity was exhibited against *Escherichia coli*, *Bacillus subtilis*, *Salmonella paratyphi* B, *Staphylococcus aureus*. As compare to Ampiclox (Table 2).

Antibacterial activity of *Raphanus sativus* Linn seeds oil:

Antibacterial activity of *Raphanus sativus* seeds oil was tested against gram +ve and

gram -ve bacteria. No antibacterial activity was exhibited against *Escherichia coil*, *Bacillus subtilis*, *Salmonella paratyphi* B and *Staphylococcus aureus*. As compare to Ampiclox (Table-3).

Antifungal activity of *Raphanus sativus* seeds oil

Preparation of samples for testing

Plant oil was not equally soluble in water, and therefore, a solution containing 100 ml of 50% ethanol and 5g of an emulsifier 'Triton' (Robin and Hass, Philadelphia, Pa 19105) was used to emulsify oil. Plant oil gives 5% concentration (i.e. 0.05g of plant oil dissolved in 1 ml solution of ethanol and triton mixture). Aliquots of this concentration were used to test antifungal activity.

Antifungal Test

The diffusion plate method was used to test plant extracts (Reddish, 1950). 0.1 ml of the fungal spore suspension (grown for 3 days in 10 ml of Sabouraud's dextrose broth) was thoroughly mixed with 25 ml of melted Sabouraud's dextrose agar and poured into

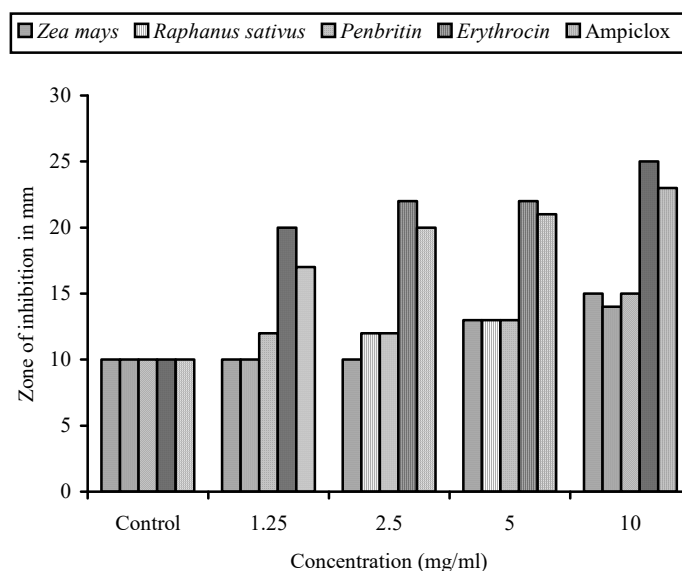


Fig. 9: Antibacterial activity against gram +ve bacteria *Bacillus pumilus*.

Zea mays Linn. (corn silk) and *Raphanus sativus* Linn. (seeds) were used in alcoholic extract.

sterilized petri-plates. When the agar was set, 5 holes of 6-mm diameter bore were made on each of the seeded plate. The four outer holes were filled with 0.2 ml of the testing sample and the central hole with 0.2 ml of 5% standard commercial fungicide "Tecto" (Merck Sharp and Dohme International, USA) for the purpose of comparing the antifungal activity of the extracts. Simultaneously another set was made with 0.2 ml of the base solution (used for dissolving plant extracts) as control in the central well of the petriplate. All these experiments were performed in duplicate. The petri-plates were incubated at 28°C for 3-5 days.

The zones of inhibition produced by plant extracts were compared with the zone produced by the standard fungicide. The (-) sign was used when no zone of inhibition was observed.

All culture plates were examined twice weekly for a period of 3 weeks and the results tabulated (Rizki *et al.*, 1987).

Antifungal activity of *Raphanus seeds oil*:

Antifungal activity of *Raphanus sativus* seeds oil was tested against *Aspergillus niger* and *Fusarium* sp. at different concentrations. Both the fungi did not show any sensitivity toward the seed oil given concentration (Table-4), while the standard Tecto checked 100% growth at 5% concentration.

RESULT AND DISCUSSION

Antibacterial activity of the alcoholic extract of *Raphanus sativus* Linn (seeds) and *Zea mays* (corn silk) against gram -ve and gram +ve bacteria is presented in Table-1 and figs.1-9. *Zea mays* (corn silk) extract was found inhibitory against *Escherichia coli* in the concentration of 02.5mg/ml whereas the minimum inhibitory concentration (MIC) for *Klebsiella pneumoniae*, *Salmonella paratyphi* A and *Salmonella typhi* was found to be 05mg/ml. *Raphanus sativus* (seeds) extract showed no activity against *Klebsiella pneumoniae* whereas the MIC against

Escherichia coli, *Salmonella paratyphi* A and *Salmonella typhi* it was 02.5 mg/ml.

Among the gram +ve bacteria studied *Zea mays* (corn silk) alcoholic extract showed no inhibition against *Staphylococcus aureus* and *Streptococcus pyogenes* but having MIC of 02.5 mg/ml against *Bacillus subtilis* and 0.5 mg/ml against *Lactobacillus plantarum* and *Bacillus pumilus*. *Raphanus sativus* (seeds) extract was found. non-inhibitory against *Staphylococcus aureus* whereas the MIC against *Lactobacillus plantarum* was found as low as 01.25 mg/ml. It inhibitory in the concentration of 02.5 mg/ml against *Staphylococcus pyogenes* and *Bacillus pumilus*. The MIC against *Bacillus subtilis* was found to be 05mg/ml.

In comparison with antibiotic *Penbritin*, *Erythrocine* and *Ampiclox*, *Zea mays* (corn silk) alcoholic extract showed greater activity against. *Klebsiella pneumoniae*, *Escherichia coli*, *Salmonella paratyphi* A and *Salmonella typhi* as compared to *Penbritin*, while *Raphanus sativus* Linn (seeds) alcoholic extract showed greater activity against all the three organism except *Klebsiella pneumoniae* as compared to *Penbritin*.

Raphanus sativus Linn (seeds) alcoholic extract and *Zea mays* (corn silk) alcoholic extract showed greater activity against *Bacillus subtilis* (gram +ve) in comparison with *Penbritin*.

Cymbopogon citratus (DC) Stapf leaves powder and *Raphanus sativus* Linn (seeds) oil did not show any antibacterial activity (Tables 2 and 3). *Raphanus sativus* Linn (seeds) oil did not show any antifungal activity (Table-4).

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